

Institute of Actuaries of India

Subject SA6 – Investment

September 2016 Examination

INDICATIVE SOLUTION

Solution 1:

i) The risks that the contract is exposed to and the possible mitigation actions are set out below:

Risk Type	Description	Mitigation Action
Investment /Reinvestment_Risk	• Future Premiums might be invested at unknown rates, especially when the yields are falling. • Coupon paying bonds are usually available in India, so the coupon reinvestment rate also in unknown. Note that the quoted GRY for any bond assumes reinvestment of coupons at the GRY itself. • Management proposed to use Corporate Bonds; these are usually shorter in tenure. Hence there is a reinvestment risk associated when these bonds mature.	• Buy fixed for floating swaps (receive fixed yields and pay floating yields). • Swaptions can also be considered – this is an option to swap sometime in the future. These are better than Swaps to the extent that if interest rates rise unpredictably high in the future, the liability of the Insurance Company would not proportionately rise (unlike in a swap mentioned above). This loss on the swaption is a maximum equal to the premium paid. In an interest rate swap the liability would be equal to the difference between the fixed and floating rates for the duration of the swap • Cost of this swap needs to be evaluated. Swap options likely to be costlier. • Counterparty risk that can arise (especially when it is an OTC swap) also needs to be considered.
Liquidity Risk (mostly related to withdrawals – might have to prematurely liquidate investments to pay for surrendering policyholders)	• Corporate Bonds are exposed to liquidity risk -if we wish to prematurely withdraw the investment it might be difficult.	• Invest only in highest rated corporate bonds whose track record for early redemptions are good.
Credit Risk	• Corporate Bonds are subject to credit risk, there could be default at an inopportune time.	• Invest only in the highest rated bonds – however this would be at the sacrifice of

		yields. • Credit Default swap (single or basket) can be used, however sufficient availability and costs could pose an issue. • Credit Spread options can also be considered where an option on the spread between the yields earned on two assets, which provides a payoff whenever the spread exceeds some level (the strike Spread). The payoff could be calculated as the difference between the value of the bond with the strike spread and the market value of the bond. • Option on Credit Default Swaps and Credit Default Swap Indices can also be considered. The latter gives protection if default occurs before the expiry date also. • OTC contracts introduce counterparty risk.
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Lastly it is worth mentioning that none of these hedging mechanisms are perfect. There could still be some residual risk which is retained by the firm. Hence it is advised to price the contract with an appropriate profit loading to represent these residual risk(s).

[10]

ii) Risk appetite

Risk appetite is often one of the most important drivers of investment policy.

The risk appetite will depend on the following factors

1. Nature of the institution

It will depend on the nature of the institution and the nature of the product within the institution's portfolio of products.

For example, some pension schemes may feel quite free to take risk within the portfolio, whereas a life insurance company may feel that a portfolio that is more closely matched to its liabilities is more appropriate.

Within a life insurance company there may be differences between portfolios that match without profit policies and those that match with profit policies due to the perceived difference in risk appetite of the policyholders.

- Level of surplus or deficit within the fund

All else being equal, a larger surplus will provide more scope for risk taking. Although sometimes a small deficit may also lead to the adoption of a risk appetite in order to grow out of the deficit.

- Marketing claims and their effect on policyholders' expectations.

The riskiness of the investment policy chosen must be demonstrably in accordance with what the policyholders have been lead to expect.

- Other factors

Indirectly, the institution's risk appetite will also be determined by:

- regulatory requirements
- accounting disclosure and treatment of surpluses and deficits

2. Size of the institution

It will also depend on the size of the institution.

This is primarily because the size of the funds available for investment will largely determine the scope for diversification of investments within the fund.

If an institution operates a very large portfolio, the level of diversification can be significant, and can reduce the overall risk to the sponsors, policyholders, shareholders and other stakeholders.

Some institutions may be so large that they are considered "too big to fail" and may find that the government would prefer to bail out policyholders rather than see them subjected to losses.

3. Term of the liabilities

Risk appetite is also influenced by the term of the liabilities. General insurers have a very short term of liabilities as compared to life insurer or pension providers. Hence the degree of freedom and choice of assets may vary significantly between these institutions due to term of liabilities.

4. Extent to which liabilities are discretionary

The extent to which liabilities are discretionary or within the control of the institution will be taken into account in determining risk appetite.

Greater discretion or control over the liabilities will generally enable more risk to be taken.

The difference between General insurance companies and pension provider will be as follows:

Since the nature of institution is different the pension provider will have longer term of liabilities than a General Insurer. Due to this Pension provider may be able to take more freedom in investment. General insurers and banks often deal with liabilities that can be quickly realized, such as annual insurance policies, or instant access accounts for banks.

General insurance companies will have very restricted risk appetite. There will also an influence on the amount of cash flow that the institutions need to generate, either through interest and dividends , or through realization of marketable investments.

The Pension Provider's investment strategy is also influenced by the sponsor covenant for a pension scheme or the Pension protection fund, or parental guarantees for subsidiary companies.

The provision of such guarantees may enable more risk-taking, as the investor is safe in the knowledge that any shortfall will be made up by the guarantor.

[10]

iii)

1. Additional Risks that arise are:
1. Customers can surrender at time when the interest rates are on the rise and consequently fixed interest securities fall in value.
2. The maximum fall could be $18\% \text{ of the value of the bonds } (\text{modified duration } (9) \times \text{expected maximum rise in yield } (2\%) = \text{expected maximum change in price})$ – so this needs to be compared with the proposed “85% of premiums paid” – to check viability from an investment risk perspective.
3. Initial expenses and commissions could further make the level of 85% of premiums paid quite onerous, especially in the initial years.
4. The surrender value offered might be unaffordable (especially in the early years) given impact of expenses etc. – hence might have to reduce the value offered.
5. Corporate Bonds may not be very liquid – hence if the fund is exposed to a large proportion of corporate bonds then untimely (and large scale) surrenders could mean that shareholders have to finance the payouts since Corporate bonds are not liquid.

[3]

iv)

The investment policy of participating funds will mainly be determined by the level of bonuses that needs to be declared for the products.

It will also depend upon the regulations that specify the asset classes in which investment can be made.

Different offices may have slightly different bonus philosophies. In general, the bigger the proportion of terminal bonus, the greater the investment freedom for the life office.

So life offices which have low guarantees and low reversionary bonus tend to hold bigger proportion of their assets in equity-type investments.

Although there will be greater investment freedom in this line of business the company will want to avoid the year on year fluctuations in the asset values. This will prevent the company in investing heavily in the risky assets.

It might also make references to the overall balance likely between real and fixed assets; give some indications of the likely overall risk profile of the portfolio, and of the type of investor who might find the risk profile appropriate to his or her investment needs.

A life insurance company is required to treat customers fairly and satisfy “policyholders’ reasonable expectations”. Arguably, it is these expectations that define the liability under a with profits policy. Reasonable expectations are likely to include:

- that the investment policy follows that stated in the company’s “principles and practices of financial management”
 - that, over the long term, the investment return should be higher than under a without profits policy
- That bonus rates are consistent with material published by the company, taking into account past bonus records.

In practice, the company will be concerned to achieve an investment return that is as good as its competitors.

Because the ability to attract new business would also depend on company investment performance and the approach towards declaration of bonuses.

[5]

v)

- a. Longevity Risk can be managed by using:
 1. Longevity Swaps – usually a swap counterparty - reinsurers or an investment Bank, involve the payment of expected pension payments on a cohort of scheme members, and the receipt of the actual pension payments required.

If actual payments rise above expected as a result of longevity, the swap will bring in sufficient net payments to the scheme to enable it to fund the payments.

The longevity risk is therefore transferred onto the swap counterparty.

Counterparty risk that arises due to this arrangement also needs to be considered.

Additional costs that occur as a result of arranging this swap also needs to be taken into account.

2. Longevity Bonds – usually issued by a large investment bank or reinsurer.

Longevity bonds pay a coupon that is proportional to the number of survivors in a selected cohort of lives.

The coupon therefore reduces each year in line with the number of survivors within the cohort and as such broadly matches the liability of the pension or annuity provider.

[4]

vi)

Cash flow matching is a method of liability hedging whereby an investor holds a portfolio of government bonds until maturity to meet a pre-specified stream of future payments. Provided the future payments do not change in amount or timing, the coupon and principal proceeds from the bond portfolio can be used to meet the obligations.

Difficulties with this approach arise for the following reasons:

- Such an approach requires a bond asset to be held that is equal in present value to the future payments discounted at bond yields (using the full yield curve). Therefore, only a partial hedge is only possible if asset cover is less than 100%.
- If the latter payments are payable after the principal payment of the longest available government bond then it will not be possible to hedge these payments at present (until longer maturity bonds become available, i.e. creating reinvestment Risk).
- Due to “gaps” between bond maturities (particularly at longer durations), there may be a need to reinvest or disinvest bonds prior to maturity, and the hedge may therefore, be imperfect.
- The use of government bonds gives risk to a (small) degree of credit risk that may not necessarily be reflected in the liability.
- If the tax status of the government bonds worsens, this will mean the assets are likely to be insufficient to meet the liability payments.
- Due to the above factors, there may be some mark to market risks between the asset value of the bond portfolio and the present value of the liability payments discounted using the bond yield curve. In some cases, this may be a material risk factor, but in other cases this will be much smaller than uncertainties in the liability payments themselves or other portfolio risks.
- This approach can be extended to payments that are subject to indexation, provided there are bonds of the appropriate maturity available with the appropriate indexation built into

their payment.

- Where there are liquid repo markets on the bonds being used to construct the liability cash flow hedge, repo contracts can be used to release funds. In this way a leveraged exposure to bonds can be created without investing the full market value.

[8]

vii)

Under the terms of an interest rate swap Insurer agrees to pay the Investment Bank (counterparty) a floating rate (typically a rate linked to the cash markets) in return for a fixed rate series of cash flows.

For inflation linked swaps the Insurer agrees to pay a fixed rate in return for inflation linked payment.

Especially for inflation linked swaps, care needs to be taken to ensure that the difference between inflationary increases for pensions in payment and the inflationary increases for the payments received under the terms of the swap are as close as possible (so this is basically a form of basis risk).

The use of swaps rather than bonds has the following advantages:

- Interest rate and inflation swap markets may have longer maturities available than bond markets.
- Swap markets may have greater liquidity and lower transaction costs than bond markets.
- Swaps permit hedging to be achieved without full asset cover being required, as they are a contract for difference rather than a funded asset.
- Swaps are in most cases bespoke contracts that are agreed with a single counterparty, rather than a standardized listed security (like a bond). Therefore greater flexibility is possible within the schedule of payments.

[4]

viii) Limitations of using swaps

- If the swaps are subject to collateralization (to mitigate credit risk), then this will require the movement and investment of collateral on a daily or weekly basis.
- The bespoke nature of a swap means that closing out a swap position is more complex than selling a bond. However, in a liquid market closing out a swap may in fact, have lower transaction costs than selling a government bond.
- Swaps are subject to counterparty risk, if the counterparty bank defaults. Whilst collateralization will limit losses, if this happens a new swap will need to be put in place at potentially higher cost (replacement risk) or the hedge lost.

- Under an interest rate swap, the receiver of the fixed interest rate will need to pay a floating interest rate to the counterparty. To the extent that there is investment risk in the assets that are used to generate the floating rate, the swap will not mitigate these risks, whereas a government bond portfolio is intrinsically low risk from a credit standpoint.
- If the swap interest rate curve moves differently to the government bond interest rate curve, this can create a basis risk, which could lead to a mark to market loss.

[3]

[47 Marks]

Solution 2:

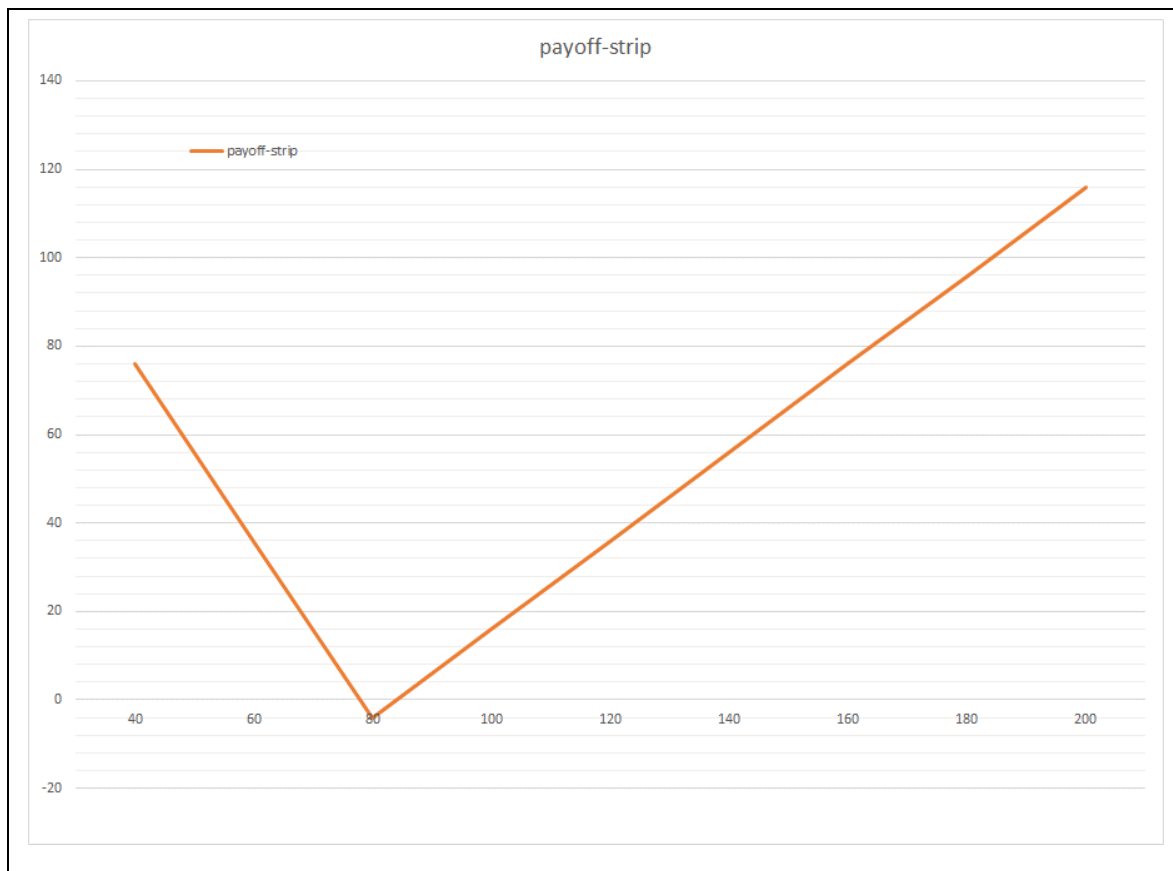
i)

Currency Risk - risk of the pound sterling rising/falling in value. This can most easily be addressed by buying Puts on the Pound Sterling so that a fall beyond a certain level is offset by profits on the Put.

Exchange traded puts are better than OTC since the latter introduces additional counterparty risks.

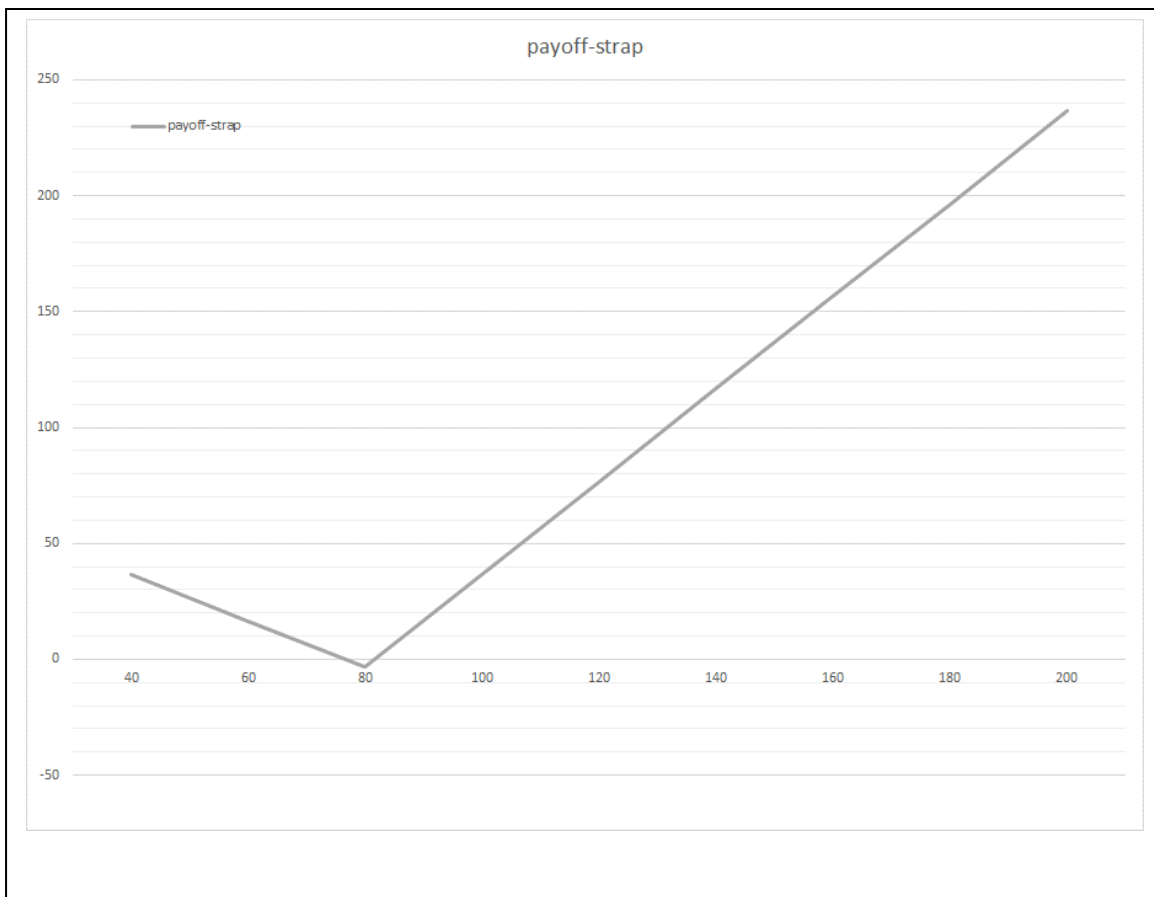
Fall is more likely than a gain in exchange rates— then the Company can buy a strip. This is essentially a combination of two puts and a call with the same exercise price. Here the Company can buy two puts for say RS.1.60 each at a strike price of INR 80 per pound (current spot price) and another call for say 90paise at the same strike price of INR 80 per pound, expiring in June.

Payoff diagram would be as below, horizontal axis represents exchange rate and vertical axis represents payoff:



Gain more likely than a fall in exchange rates – then the Company can buy a strap. This is essentially a long position in two calls and one put with the same exercise price and expiry date.

Here the Company can buy two calls for say 90 paise each at a strike price of INR 80 per pound (current spot price) and another put for say INR 1.60 at the same strike price of INR 80 per pound, expiring in June.



[10]

ii)

- a. The exposure to credit risk can be hedged by buying credit default swaps directly - either buy a CDS on a basket of entities (here the underlying is a basket of entities) or buy a Credit Default Index swap (which is a portfolio of single entity CDS').

However, a better way to hedge the credit risk in the current situation is to buy options on the Credit Default Index Swaps (CDIS). Options on CDS where underlying is a basket of entities would not be as effective as the former if default occurs before expiry.

Buying options may at times be cheaper and can give the required protection should the credit event occur. When an option is bought on a CDIS, the protection seller compensates the protection buyer on exercise of the option should the credit event occur.

Options on CDS where underlying is a basket of entities, does not offer any protection if default occurs prior to expiry of option, unlike an option on an Index as explained above .

Especially given that this is a large investment house, the use of option would prove more economical. Also it is not sure whether or not UK would exit the EU and if it does, the impact on credit ratings of sterling denominated bonds. So buying options is recommended.

Given that this is a large investment house, it is expected that the portfolio of corporate bonds should be well diversified and hence an option on a Credit Default Swap Index would be appropriate.

[6]

iii)

a) Mid-cap versus Large Cap

Mid-Cap companies (part of FTSE-250) are more reliant on the domestic economy and consumer spending than the large, multinational companies that dominate the FTSE 100, who may even benefit from the post-Brexit slump in sterling as a weaker pound makes their exports more affordable overseas.

E.g. Property shares (which in turn will hit domestically focused British banks) and domestic retailers etc. are expected to be more vulnerable to a fall post Brexit.

[3]

b) Strategy to be adopted

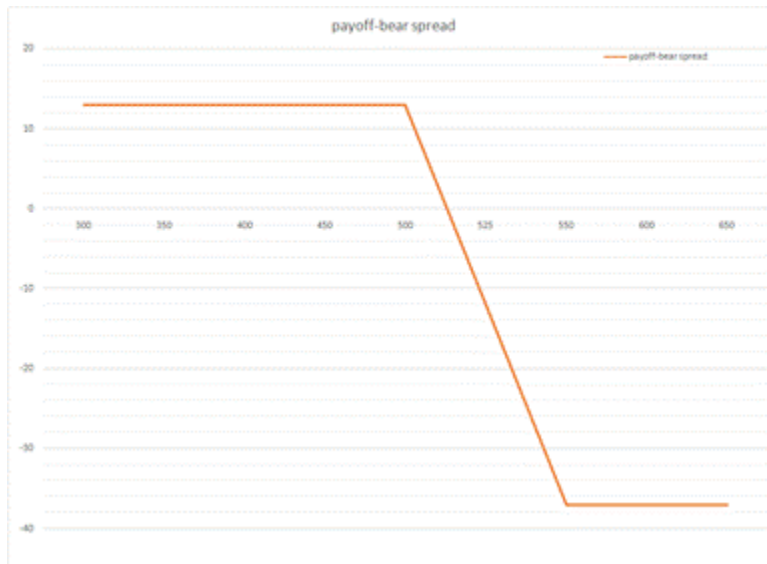
Since the expectation of the Brexit vote is negative (falling asset prices and weakening pound) – then we could profit more from creating derivative strategies on mid-cap stocks (or the FTSE 250 index itself).

A bear spread is can be used since we are concerned with downside risk and wish to profit from the same.

We can buy a call (or put) with a higher strike price and sell a call (or put) with a lower strike price, each option having the same expiry date. This will create a payoff profile that is profitable for the investor if the share price falls.

Consider a bear spread on a chosen stock as follows: buy a 550 call for 47p and sell a 500 call for 60p

Payoff Diagram



Other plausible answers include (along with suitable payoff diagram and example). :

1. Short sell index futures and buy back after the fall
2. Buy puts and profit from a fall. The profits will be reduced by the put option premium paid upfront at the time of purchase.
3. Sell call options and profit from the fall. Profits would be just the option premium received upfront.

[6]

iv)

a) What is QE

Quantitative Easing (QE) is a monetary policy used by some central banks to increase the supply of money. It usually involves a direct increase in the money supply or a reduction in the CRR (Cash Reserve Ratio) or SLR (Statutory Liquidity Ratio).

QE is usually implemented by a central bank by first crediting its own account with money it creates ex nihilo ("out of nothing"). It then purchases financial assets, for example government bonds, agency debt, mortgage-backed securities and corporate bonds, from banks and other financial institutions in a process referred to as open market operations. It can also involve changing the reserve requirements would increase the money supply

[4]

b) Impact

Fixed interest government bonds

The short term interest rates would see a decline leading to an increase in the price of the short term bonds. The effect of changes in the short term interest rates on yields of longer term bonds is not as clear-cut. According to Expectations theory if the market perceives that the future short term interest rates will also remain low as per the decline in the current short term rates then the long yields would fall and the bond prices for longer duration bonds in the portfolio would also rise.

However on the contrary if the market perceives that the quantitative easing would lead to high inflation in the future then the long term yields would not reduce as much as the short term ones and may even rise.

The bond portfolio will also be affected if QE in other parts of the world affects the yields of other international government bonds, as they are comparable to bonds in the UK.

Therefore, the longer global QE is perceived to continue, the longer the low yields on long bonds continue.

On the other hand, QE can cause the currency to weaken due to the low interest rates...

... which in turn can cause imported raw material prices and input costs for companies to rise ...

... which can create cost push inflation in the economy.

Inflation will also occur as a result of the increase in the money supply through the printing of money by the central bank.

Also, because QE increases the cash reserves that commercial banks have with the central bank, the money multiplier effect means that commercial banks are free to expand their lending books.

If commercial banks extend new loans to the public and to companies in the form of mortgage finance or corporate loans, this may cause demand pull inflation in the economy.

If inflation pressures build, then long government bond yields will gradually rise.

This is because, according to the inflation risk theory, the yield on a government bond should be linked to the required return which is:

risk-free real return + expected inflation + inflation risk premium

Index-linked government bonds

Index-linked bonds are immune to inflationary pressures – the prices are determined by the risk-free real return.

However, if the QE pushes the prices of fixed coupon bonds to artificially high levels, there is a link between the return on index-linked and the return on fixed government bonds, so index linked prices would also be pushed upwards.

Real yields can (and have) become negative.

Also as a hedge against inflation, they might benefit from a flight to quality as investors seek assets that are safe in the uncertain economic environment.

Domestic equity shares

QE makes the currency weak in the international markets, which will make exporters more competitive and increase their profits.

It will however increase raw material and input costs which will hurt domestic companies that cannot benefit from export competitiveness.

If bond yields are pushed to artificially low levels in the short term, investors will look for alternatives that offer a reasonable income yield, so equities might benefit.

Given that equities should offer the prospect of capital growth, they may be seen as a cheap alternative to gilts.

Companies that have long and medium-term loans will benefit from lower borrowing costs and profits should improve.

Lower rates should reduce the cost of any new corporate bond issues which should encourage investment and benefit growth.

Improved consumer confidence and improved availability of loan finance should increase economic growth and benefit equities.

[12]

v)

a) Financial repression consists of the following:

1. Explicit or indirect capping of interest rates, such as on government debt and deposit rates (0.5)
 2. Government ownership or control of domestic banks and financial institutions with barriers that limit other institutions from entering the market.
 3. High reserve (SLR) requirements
 4. Creation or maintenance of a captive domestic market for government debt, achieved by requiring banks to hold government debt via capital requirements, or by prohibiting or disincentives alternatives.
 5. Government restrictions on the transfer of assets abroad through the imposition of capital controls.
- These measures allow governments to issue debt at lower interest rates.

[2]

b) There are two main implications of falling interest rates on the Pension funds.

- Scheme Funding implications and impact on sponsors
- Individual annuity prices and fund values

In particular, for Defined benefit schemes there may be following impact:

There might be higher recovery payments for sponsor at a time when it may find hardest to afford.

Possible weakening of on-going business prospects and/or jeopardize ability to continue to support the scheme.

Scheme Trustees are faced with higher gilt prices, making it costly to pursue a matched strategy.

For the defined contribution scheme there would be following impact:

Increased annuity purchase price due to the low interest rates

Higher contribution may be required to be able to target a similar level of expected annuity payouts

Impact may be dampened for those very close to retirement.

Low or negative investment returns, especially if purchasing index linked gilts.

[6]

c) Life Insurance:

- Inflated gilt prices may push investment strategy towards alternate investments in search of yield. Investment in corporate bonds and structured credit is more attractive to maintain rates of return.
- Possible balance sheet impacts
 - Higher market risk capital requirements if alternative investment strategy pursued.
 - Lower gilt yield may lead to lower reserving interest rate and thus higher reserves impacting the balance sheet position (greatest impact for deferred products).
 - It may also lead to revision in the pricing interest rates and refiling of products.
 - Cost of annuity rate and investment guarantees increased.
 - Higher longevity capital requirement.
- Fall in perceived value of saving may lead to less saving by individuals and consequently less business.

[4]**[53 Marks]**
